Binomial thinning credibility model for total and extreme claim counts

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Abstract

To be written.

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1. Introduction

In many insurance portfolios, two claim counts are recorded for the same risk unit and period: the total number of claims and the number of extreme claims among them, for example the claims whose amounts exceed a large-loss threshold. The two counts are nested, since the number of extreme claims cannot exceed the total number of claims, and they are observed repeatedly over time. A joint model for such data should respect the nesting, accommodate time-varying covariates, and allow the past experience of a risk unit to update the predictions of both counts.

This paper proposes a model in which the extreme claim count is a binomial thinning of the total claim count. The total claim count of a region follows a Poisson distribution with a gamma random effect, and each claim is classified as extreme with a probit probability that depends on a Gaussian random effect. Under independent and static random effects, we show that the likelihood factorizes into a Poisson-gamma part and a Bernoulli-probit part, so that the two sets of parameters can be estimated separately. We further derive the posterior distributions of the random effects and the exact Bayesian one-step predictions of both counts in closed form. We illustrate the model with annual claim counts of 17 Korean regions over 2015–2024 and with the claim counts of about 1,200 local government entities in Wisconsin over 2006–2011.

The paper is organized as follows. Section 2 introduces the notation and two auxiliary results on Gaussian random effects in probit models. Section 3 presents the model and the main results. Section 4 applies the model to a regional panel of total and extreme claim counts, and Section 5 applies it to an entity-level panel of total and vehicle claim counts from the Wisconsin Local Government Property Insurance Fund.

2. Preliminaries

2.1. Notation

Let $\mathbb{N}$ denote the set of positive integers and $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$. We write $\mathbb{R}$, $\mathbb{R}_{\geq 0}$, and $\mathbb{R}_{>0}$ for the real line, the nonnegative real line, and the positive real line, respectively. For $k \in \mathbb{N}$, vectors in $\mathbb{R}^k$ are column vectors, and we write $x_{1:k} := (x_1, \ldots, x_k)^\top$. The $k \times k$ identity matrix is denoted by $\boldsymbol{I}_k$. Inequalities between vectors are understood componentwise. A product over an empty index set equals 1, and a sum over an empty index set equals 0.

We use the following distributions.

- Gamma(a, b): gamma distribution with shape $a \in \mathbb{R}_{>0}$ and rate $b \in \mathbb{R}_{>0}$, so that its mean is a/b.
- N(μ, σ^2): normal distribution with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 \in \mathbb{R}_{>0}$.
- MVN$_k(\mu_{1:k}, \boldsymbol{\Sigma})$: k-variate normal distribution with mean $\mu_{1:k} \in \mathbb{R}^k$ and positive semidefinite covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{k \times k}$.
- Pois(λ): Poisson distribution with mean $\lambda \in \mathbb{R}_{>0}$.

- $\mathrm{Bin}(n, p)$: binomial distribution with $n \in \mathbb{N}_0$ trials and success probability $p \in (0, 1)$, where $\mathrm{Bin}(0, p)$ is the point mass at 0.

Let ϕ and Φ denote the density and the distribution function of $\mathrm{N}(0, 1)$. For a positive semidefinite matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{k \times k}$, let

$$\Phi_k(x_{1:k}; \boldsymbol{\Sigma}) := \mathbb{P}\left(X_{1:k} \leq x_{1:k}\right), \qquad X_{1:k} \sim \mathrm{MVN}_k(0_{1:k}, \boldsymbol{\Sigma}), \quad x_{1:k} \in \mathbb{R}^k,$$

where $0_{1:k} \in \mathbb{R}^k$ is the zero vector. For $k = 0$, we set $\Phi_0 := 1$.

2.2. Gaussian random effects in probit models

The following identity converts the expectation of a product of probit functions of a Gaussian vector into a multivariate normal distribution function.

Lemma 1 (Gaussian product-probit identity)**.** *Let $X_{1:k} \sim \mathrm{MVN}_k(\mu_{1:k}, \boldsymbol{\Sigma})$, where $\mu_{1:k} \in \mathbb{R}^k$ and $\boldsymbol{\Sigma} \in \mathbb{R}^{k \times k}$ is positive semidefinite. Then*

$$\mathbb{E}\left[\prod_{j=1}^{k} \Phi(X_j)\right] = \Phi_k\left(\mu_{1:k}; \boldsymbol{I}_k + \boldsymbol{\Sigma}\right).$$

Proof. Let $W_{1:k} \sim \mathrm{MVN}_k(0_{1:k}, \boldsymbol{I}_k)$ be independent of $X_{1:k}$. Since $W_1, \ldots, W_k$ are independent standard normal random variables,

$$\prod_{j=1}^{k} \Phi(X_j) = \mathbb{P}\left(W_{1:k} \leq X_{1:k} \mid X_{1:k}\right).$$

Taking expectations gives

$$\mathbb{E}\left[\prod_{j=1}^{k} \Phi(X_j)\right] = \mathbb{P}\left(W_{1:k} \leq X_{1:k}\right) = \mathbb{P}\left(W_{1:k} - (X_{1:k} - \mu_{1:k}) \leq \mu_{1:k}\right).$$

Since $W_{1:k}$ and $X_{1:k}$ are independent Gaussian vectors,

$$W_{1:k} - (X_{1:k} - \mu_{1:k}) \sim \mathrm{MVN}_k\left(0_{1:k}, \boldsymbol{I}_k + \boldsymbol{\Sigma}\right),$$

which proves the result. □

Lemma 1 does not require $\boldsymbol{\Sigma}$ to be positive definite. This is essential for the next result, in which a single scalar random effect is shared by all components, so that the covariance matrix has rank one.

Lemma 2 (Bernoulli-probit random effect)**.** *Let* $\Theta \sim \mathrm{N}(0, \sigma^2)$ *with* $\sigma > 0$*, and let* $k \in \mathbb{N}$*,* $u_{1:k} \in \{-1, 1\}^k$*, and* $h_{1:k} \in \mathbb{R}^k$*. Then*

$$\begin{aligned}\mathbb{E}\left[\prod_{j=1}^{k} \Phi\left(u_j(h_j + \Theta)\right)\right] &= \int_{\mathbb{R}} \phi(z) \prod_{j=1}^{k} \Phi\left(u_j(h_j + \sigma z)\right) \mathrm{d}z \\ &= \Phi_k\left((u_1 h_1, \ldots, u_k h_k)^\top; \boldsymbol{I}_k + \sigma^2 u_{1:k} u_{1:k}^\top\right).\end{aligned}$$

Proof. The first equality follows since Θ has the same distribution as σZ with $Z \sim \mathrm{N}(0, 1)$. For the second equality, write

$$X_{1:k} := (u_1(h_1 + \Theta), \ldots, u_k(h_k + \Theta))^\top = (u_1 h_1, \ldots, u_k h_k)^\top + u_{1:k}\Theta.$$

Then $X_{1:k} \sim \mathrm{MVN}_k\left((u_1 h_1, \ldots, u_k h_k)^\top, \sigma^2 u_{1:k} u_{1:k}^\top\right)$, and Lemma 1 proves the result. □

Remark 1. *Lemma 2 has the following interpretation. Let* $Z_1, \ldots, Z_k$ *be conditionally independent given* Θ *with* $Z_j \mid \Theta \sim \mathrm{Bin}\left(1, \Phi(h_j + \Theta)\right)$*. Since* $1 - \Phi(a) = \Phi(-a)$ *for* $a \in \mathbb{R}$*, we have* $\mathbb{P}\left(Z_j = z_j \mid \Theta\right) = \Phi\left(u_j(h_j + \Theta)\right)$ *with* $u_j := 2z_j - 1$ *for* $z_j \in \{0, 1\}$*. Hence Lemma 2 gives the joint probability* $\mathbb{P}\left(Z_{1:k} = z_{1:k}\right)$ *in closed form. The covariance matrix* $\sigma^2 u_{1:k} u_{1:k}^\top$ *is of rank one, reflecting the single random effect shared by all trials, and the one-dimensional integral representation is convenient for computation when* k *is large.*

3. Binomial thinning model for bivariate claim frequency

In this section, we develop a model for two nested claim counts. For each region i and period t, let N_{it} denote the total number of claims and let M_{it} denote the number of extreme claims among them, for example, the claims whose amounts exceed a given threshold. By construction,

$$0 \le M_{it} \le N_{it}.$$

A natural way to respect this constraint is to regard M_{it} as a binomial thinning of N_{it}: each of the N_{it} claims is classified as extreme or not. Repeated observations of the same region over time are linked through two region-specific random effects, one for the claim frequency and one for the probability that a claim is extreme. Throughout this section, the two random effects are assumed to be independent and static over time, and predictions are obtained as exact Bayesian posterior means.

Model 1 (Poisson-gamma binomial-probit thinning model)**.** *Let*

$$r, \sigma \in \mathbb{R}_{>0}, \qquad \alpha, \beta \in \mathbb{R}^p,$$

and let $x_{it} \in \mathbb{R}^p$*,* $i = 1, \ldots, I$*,* $t \in \mathbb{N}$*, be predetermined exogenous covariates, which are allowed to*

vary over time. Define

$$\lambda_{it} := \exp\left(x_{it}^\top \alpha\right) \in \mathbb{R}_{>0}, \qquad \eta_{it} := x_{it}^\top \beta \in \mathbb{R}.$$

We define the bivariate sequences $((N_{it}, M_{it}))_{t\in\mathbb{N}}$, $i = 1, \ldots, I$, *through the following random effect assumption and observation equations.*

i. Random effects: For each region i, *assume*

$$\Theta_i^{(1)} \sim \text{Gamma}(r, r), \qquad \Theta_i^{(2)} \sim \text{N}(0, \sigma^2),$$

where $\Theta_i^{(1)}$ *and* $\Theta_i^{(2)}$ *are independent. Furthermore, the pairs* $\left(\Theta_i^{(1)}, \Theta_i^{(2)}\right)$, $i = 1, \ldots, I$, *are mutually independent.*

ii. Observation equations: Conditional on $\left(\Theta_i^{(1)}, \Theta_i^{(2)}\right)$, *assume that the pairs* (N_{it}, M_{it}), $t \in \mathbb{N}$, *are mutually independent with*

$$N_{it} \mid \Theta_i^{(1)}, \Theta_i^{(2)} \sim \text{Pois}\left(\lambda_{it}\Theta_i^{(1)}\right)$$

and

$$M_{it} \mid N_{it}, \Theta_i^{(1)}, \Theta_i^{(2)} \sim \text{Bin}\left(N_{it}, \Phi\left(\eta_{it} + \Theta_i^{(2)}\right)\right),$$

Furthermore, the regions are independent: the sequences $((N_{it}, M_{it}))_{t\in\mathbb{N}}$, $i = 1, \ldots, I$, *are mutually independent.*

Here, $\text{Bin}(0, p)$ is the point mass at 0, so that $M_{it} = 0$ whenever $N_{it} = 0$.

Since $\mathbb{E}\left[\Theta_i^{(1)}\right] = 1$ and $\mathbb{E}\left[\Theta_i^{(2)}\right] = 0$, the intercepts in α and β are identifiable, and λ_{it} is the a priori expected total claim count. The random effect $\Theta_i^{(1)}$ captures the unobserved heterogeneity of the claim frequency across regions, whereas $\Theta_i^{(2)}$ captures the unobserved heterogeneity of the propensity of a claim to be extreme. For notational convenience, write

$$p_{it}(\theta) := \Phi\left(\eta_{it} + \theta\right), \qquad \theta \in \mathbb{R}.$$

Remark 2. *All results in this section use* λ_{it} *and* η_{it} *only as known constants. Hence they remain valid for any* $\lambda_{it} \in \mathbb{R}_{>0}$ *and* $\eta_{it} \in \mathbb{R}$*; in particular, an exposure offset* $e_{it} > 0$ *can be included through* $\lambda_{it} = e_{it} \exp\left(x_{it}^\top \alpha\right)$.

The following lemma is the well-known thinning property of the Poisson distribution. It shows that, conditional on the random effects, the extreme and non-extreme claim counts are independent Poisson random variables.

Lemma 3 (Poisson thinning)**.** *Under Model 1, conditional on* $\left(\Theta_i^{(1)}, \Theta_i^{(2)}\right)$, *the random variables* M_{it} *and* $N_{it} - M_{it}$ *are independent with*

$$M_{it} \mid \Theta_i^{(1)}, \Theta_i^{(2)} \sim \text{Pois}\left(\lambda_{it}\Theta_i^{(1)} p_{it}\left(\Theta_i^{(2)}\right)\right)$$

and

$$N_{it} - M_{it} \mid \Theta_i^{(1)}, \Theta_i^{(2)} \sim \mathrm{Pois}\left(\lambda_{it}\Theta_i^{(1)}\left\{1 - p_{it}\left(\Theta_i^{(2)}\right)\right\}\right).$$

Proof. Fix $\theta_1 > 0$ and $\theta_2 \in \mathbb{R}$, and write $\Lambda := \lambda_{it}\theta_1$ and $p := p_{it}(\theta_2)$. For $m, k \in \mathbb{N}_0$, Model 1 gives

$$\begin{aligned}
&\mathbb{P}\left(M_{it} = m,\, N_{it} - M_{it} = k \mid \Theta_i^{(1)} = \theta_1, \Theta_i^{(2)} = \theta_2\right) \\
&\quad = \mathbb{P}\left(N_{it} = m + k \mid \Theta_i^{(1)} = \theta_1\right)\binom{m+k}{m} p^m (1-p)^k \\
&\quad = \frac{e^{-\Lambda}\Lambda^{m+k}}{(m+k)!}\frac{(m+k)!}{m!\,k!} p^m (1-p)^k \\
&\quad = \frac{e^{-\Lambda p}(\Lambda p)^m}{m!} \cdot \frac{e^{-\Lambda(1-p)}\{\Lambda(1-p)\}^k}{k!},
\end{aligned}$$

which proves the result. □

Since the regions are independent, it suffices to study a single region. The following remark collects the notation for its observed history.

Remark 3. *Fix a region i and $t \in \mathbb{N}$, and consider a realized history*

$$\mathcal{H}_{it} := \{(N_{ij}, M_{ij}) = (n_{ij}, m_{ij}) :\ j = 1, \dots, t\}, \qquad n_{ij}, m_{ij} \in \mathbb{N}_0, \quad m_{ij} \le n_{ij}.$$

Define the cumulative claim count and the cumulative a priori frequency by

$$n_{i\bullet t} := \sum_{j=1}^{t} n_{ij}, \qquad \lambda_{i\bullet t} := \sum_{j=1}^{t} \lambda_{ij}.$$

Each of the $n_{i\bullet t}$ observed claims is regarded as a Bernoulli trial, which is a success if the claim is extreme. To this end, for $j = 1, \dots, t$, define the sign vector

$$u^{(j)} := \big(\underbrace{1, \dots, 1}_{m_{ij}}, \underbrace{-1, \dots, -1}_{n_{ij} - m_{ij}}\big)^\top \in \{-1, 1\}^{n_{ij}},$$

which is empty when $n_{ij} = 0$, and stack these vectors as

$$u_{1:n_{i\bullet t}} := \left(u^{(1)\top}, \dots, u^{(t)\top}\right)^\top, \qquad h_{1:n_{i\bullet t}} := \big(\underbrace{\eta_{i1}, \dots, \eta_{i1}}_{n_{i1}}, \dots, \underbrace{\eta_{it}, \dots, \eta_{it}}_{n_{it}}\big)^\top.$$

That is, u_k is the sign of the kth claim and h_k is the value of η_{ij} in the period j of the kth claim. We then define

$$\mu_{it} := (u_1 h_1, \dots, u_{n_{i\bullet t}} h_{n_{i\bullet t}})^\top \in \mathbb{R}^{n_{i\bullet t}}, \qquad \boldsymbol{U}_{it} := u_{1:n_{i\bullet t}} u_{1:n_{i\bullet t}}^\top \in \mathbb{R}^{n_{i\bullet t} \times n_{i\bullet t}},$$

which are the mean vector and the rank-one matrix appearing in Lemma 2 for the trials formed by the observed claims.

For $n, m \in \mathbb{N}_0$ with $m \le n$, let $\mu_{it}^{(n,m)} \in \mathbb{R}^{n_{i\bullet t}+n}$ and $\boldsymbol{U}_{it}^{(n,m)} \in \mathbb{R}^{(n_{i\bullet t}+n)\times(n_{i\bullet t}+n)}$ denote the corresponding quantities for the augmented history $\mathcal{H}_{it} \cup \{(N_{i,t+1}, M_{i,t+1}) = (n, m)\}$. That is, n additional trials with η-value $\eta_{i,t+1}$ are appended, m of them with sign 1 and $n - m$ of them with sign -1.

Finally, define

$$D_{it}(\theta) := \prod_{j=1}^{t} p_{ij}(\theta)^{m_{ij}} \left\{1 - p_{ij}(\theta)\right\}^{n_{ij}-m_{ij}}, \qquad \theta \in \mathbb{R}.$$

Since $1 - \Phi(a) = \Phi(-a)$ for $a \in \mathbb{R}$, we have

$$D_{it}(\theta) = \prod_{k=1}^{n_{i\bullet t}} \Phi\left(u_k(h_k + \theta)\right). \tag{3.1}$$

For $n_{i\bullet t} = 0$, we adopt the conventions $D_{it}(\theta) = 1$ and $\Phi_0\left(\mu_{it}; \boldsymbol{I}_0 + \sigma^2 \boldsymbol{U}_{it}\right) = 1$.

The following theorem shows that the joint probability of the observed history factorizes into a closed-form Poisson-gamma part, which depends only on (α, r), and a Bernoulli-probit part of the form in Lemma 2, which depends only on (β, σ).

Theorem 1 (Likelihood factorization)**.** *Consider Model 1 with the realized history $\mathcal{H}_{it}$, and let the notation be as in Remark 3. Then*

$$\mathbb{P}\left(\mathcal{H}_{it}\right) = L_{it}^{(1)}\, L_{it}^{(2)}, \tag{3.2}$$

where

$$L_{it}^{(1)} := \mathbb{P}\left(N_{i,1:t} = n_{i,1:t}\right) = \frac{\Gamma\left(r + n_{i\bullet t}\right)}{\Gamma(r) \prod_{j=1}^{t} n_{ij}!} \frac{r^r \prod_{j=1}^{t} \lambda_{ij}^{n_{ij}}}{\left(r + \lambda_{i\bullet t}\right)^{r+n_{i\bullet t}}} \tag{3.3}$$

and

$$\begin{aligned} L_{it}^{(2)} &:= \mathbb{P}\left(M_{i,1:t} = m_{i,1:t} \mid N_{i,1:t} = n_{i,1:t}\right) \\ &= \prod_{j=1}^{t} \binom{n_{ij}}{m_{ij}} \int_{\mathbb{R}} \frac{1}{\sigma} \phi\left(\frac{\theta}{\sigma}\right) D_{it}(\theta)\, \mathrm{d}\theta \\ &= \prod_{j=1}^{t} \binom{n_{ij}}{m_{ij}} \Phi_{n_{i\bullet t}}\left(\mu_{it}; \boldsymbol{I}_{n_{i\bullet t}} + \sigma^2 \boldsymbol{U}_{it}\right). \end{aligned} \tag{3.4}$$

Proof. By the conditional independence in Model 1,

$$\mathbb{P}\left(\mathcal{H}_{it}\right) = \mathbb{E}\left[\prod_{j=1}^{t} \mathbb{P}\left(N_{ij} = n_{ij} \mid \Theta_i^{(1)}\right) \mathbb{P}\left(M_{ij} = m_{ij} \mid N_{ij} = n_{ij}, \Theta_i^{(2)}\right)\right].$$

The first factor in the product depends only on $\Theta_i^{(1)}$ and the second only on $\Theta_i^{(2)}$. Since $\Theta_i^{(1)}$ and

$\Theta_i^{(2)}$ are independent, we obtain

$$\mathbb{P}\left(\mathcal{H}_{it}\right) = \mathbb{E}\left[\prod_{j=1}^{t} \mathbb{P}\left(N_{ij} = n_{ij} \mid \Theta_i^{(1)}\right)\right] \mathbb{E}\left[\prod_{j=1}^{t} \mathbb{P}\left(M_{ij} = m_{ij} \mid N_{ij} = n_{ij}, \Theta_i^{(2)}\right)\right].$$

For the first expectation, the gamma integral gives

$$\begin{aligned}\mathbb{E}\left[\prod_{j=1}^{t} \mathbb{P}\left(N_{ij} = n_{ij} \mid \Theta_i^{(1)}\right)\right] &= \int_0^\infty \prod_{j=1}^{t} \frac{e^{-\lambda_{ij}\theta}(\lambda_{ij}\theta)^{n_{ij}}}{n_{ij}!} \frac{r^r}{\Gamma(r)} \theta^{r-1} e^{-r\theta} \,\mathrm{d}\theta \\ &= \frac{r^r \prod_{j=1}^{t} \lambda_{ij}^{n_{ij}}}{\Gamma(r) \prod_{j=1}^{t} n_{ij}!} \int_0^\infty \theta^{r+n_{i\bullet t}-1} e^{-(r+\lambda_{i\bullet t})\theta} \,\mathrm{d}\theta,\end{aligned}$$

which equals the right-hand side of (3.3). Summing the first display over all $m_{i,1:t}$ with $0 \le m_{ij} \le n_{ij}$ shows that this expectation is $\mathbb{P}\left(N_{i,1:t} = n_{i,1:t}\right)$, because the second expectation sums to one.

For the second expectation, we have

$$\mathbb{E}\left[\prod_{j=1}^{t} \mathbb{P}\left(M_{ij} = m_{ij} \mid N_{ij} = n_{ij}, \Theta_i^{(2)}\right)\right] = \prod_{j=1}^{t} \binom{n_{ij}}{m_{ij}} \mathbb{E}\left[D_{it}\left(\Theta_i^{(2)}\right)\right],$$

which proves the second line of (3.4). It then follows from (3.2) and (3.3) that this expression is $\mathbb{P}\left(M_{i,1:t} = m_{i,1:t} \mid N_{i,1:t} = n_{i,1:t}\right)$. Finally, by (3.1),

$$\mathbb{E}\left[D_{it}\left(\Theta_i^{(2)}\right)\right] = \mathbb{E}\left[\prod_{k=1}^{n_{i\bullet t}} \Phi\left(u_k\left(h_k + \Theta_i^{(2)}\right)\right)\right],$$

and applying Lemma 2 with $k = n_{i\bullet t}$, $u_{1:k}$, and $h_{1:k}$ proves the last line of (3.4). □

Remark 4 (Separable estimation)**.** *By Theorem 1 and the independence of the regions, the log-likelihood of the observations $\mathcal{H}_{1T}, \ldots, \mathcal{H}_{IT}$ decomposes as*

$$\ell(\alpha, r, \beta, \sigma) = \ell^{(1)}(\alpha, r) + \ell^{(2)}(\beta, \sigma), \qquad \ell^{(k)} := \sum_{i=1}^{I} \log L_{iT}^{(k)}, \quad k = 1, 2.$$

Hence the maximum likelihood estimators of (α, r) and (β, σ) are obtained from two separate optimizations, and the Fisher information matrix is block-diagonal. Here, $\ell^{(1)}$ is the log-likelihood of a negative binomial random effect panel model for N_{it}, and $\ell^{(2)}$ is the log-likelihood of a binomial probit model for M_{it} given N_{it} with a Gaussian random intercept. Periods with $n_{it} = 0$ do not contribute to $\ell^{(2)}$.

Remark 5 (Computation)**.** *The dimension $n_{i\bullet t}$ of the multivariate normal distribution function in (3.4) equals the cumulative number of claims of region i, which can be large. However, since $\boldsymbol{U}_{it}$*

is of rank one, the second line of (3.4) expresses the same quantity as the one-dimensional integral

$$\Phi_{n_{i\bullet t}}\left(\mu_{it}; \boldsymbol{I}_{n_{i\bullet t}} + \sigma^2 \boldsymbol{U}_{it}\right) = \int_{\mathbb{R}} \phi(z)\, D_{it}(\sigma z)\, \mathrm{d}z,$$

which depends on the data only through the t pairs (n_{ij}, m_{ij}) and can be evaluated accurately by Gauss–Hermite quadrature.

The following corollary shows that the two random effects remain independent a posteriori, and that the frequency random effect has a conjugate gamma posterior.

Corollary 1 (Posterior distribution of the random effects)**.** *Under the setting of Theorem 1, the following results hold.*

i. Conditional on $\mathcal{H}_{it}$, the random effects $\Theta_i^{(1)}$ and $\Theta_i^{(2)}$ are independent.

ii. We have

$$\Theta_i^{(1)} \mid \mathcal{H}_{it} \sim \mathrm{Gamma}\left(r + n_{i\bullet t},\, r + \lambda_{i\bullet t}\right).$$

iii. The conditional distribution $\Theta_i^{(2)} \mid \mathcal{H}_{it}$ has density $\pi(\theta \mid \mathcal{H}_{it})$ at $\theta \in \mathbb{R}$ given by

$$\pi(\theta \mid \mathcal{H}_{it}) = \frac{\dfrac{1}{\sigma}\phi\left(\dfrac{\theta}{\sigma}\right) \displaystyle\prod_{k=1}^{n_{i\bullet t}} \Phi\left(u_k(h_k + \theta)\right)}{\Phi_{n_{i\bullet t}}\left(\mu_{it}; \boldsymbol{I}_{n_{i\bullet t}} + \sigma^2 \boldsymbol{U}_{it}\right)}. \tag{3.5}$$

Proof. Let π_1 and π_2 denote the prior densities of $\Theta_i^{(1)}$ and $\Theta_i^{(2)}$, respectively. By Bayes' formula and the proof of Theorem 1, the joint posterior density of $\left(\Theta_i^{(1)}, \Theta_i^{(2)}\right)$ at $(\theta_1, \theta_2) \in \mathbb{R}_{>0} \times \mathbb{R}$ is

$$\frac{\pi_1(\theta_1) \prod_{j=1}^{t} \mathbb{P}\left(N_{ij} = n_{ij} \mid \Theta_i^{(1)} = \theta_1\right)}{L_{it}^{(1)}} \cdot \frac{\pi_2(\theta_2) \prod_{j=1}^{t} \binom{n_{ij}}{m_{ij}} D_{it}(\theta_2)}{L_{it}^{(2)}}.$$

Each factor integrates to one, which proves part i. The first factor is proportional to

$$\theta_1^{r + n_{i\bullet t} - 1} e^{-(r + \lambda_{i\bullet t})\theta_1},$$

which proves part ii. Substituting (3.1) and (3.4) into the second factor proves part iii. □

By part ii of Corollary 1, the posterior of the frequency random effect depends on the history only through $n_{i\bullet t}$; in particular, the extreme claim counts $m_{i,1:t}$ carry no information on $\Theta_i^{(1)}$. Conversely, by part iii, the total claim counts affect the posterior of $\Theta_i^{(2)}$ only by determining the number of Bernoulli trials. The density (3.5) is a Gaussian density multiplied by a product of probit functions, normalized by the multivariate normal distribution function in Lemma 2.

We now present the exact Bayesian one-step-ahead predictions.

Theorem 2 (Closed-form one-step prediction)**.** *Consider Model 1 with the realized history $\mathcal{H}_{it}$, and let the notation be as in Remark 3. Then, we have the following results.*

i. For $n, m \in \mathbb{N}_0$ with $m \leq n$, the predictive probability is

$$\begin{aligned}
&\mathbb{P}\left(N_{i,t+1} = n,\, M_{i,t+1} = m \mid \mathcal{H}_{it}\right) \\
&\quad = \frac{\Gamma\left(r + n_{i\bullet t} + n\right)}{\Gamma\left(r + n_{i\bullet t}\right) n!} \left(\frac{r + \lambda_{i\bullet t}}{r + \lambda_{i\bullet t} + \lambda_{i,t+1}}\right)^{r+n_{i\bullet t}} \left(\frac{\lambda_{i,t+1}}{r + \lambda_{i\bullet t} + \lambda_{i,t+1}}\right)^{n} \\
&\qquad \times \binom{n}{m} \frac{\Phi_{n_{i\bullet t}+n}\left(\mu_{it}^{(n,m)}; \boldsymbol{I}_{n_{i\bullet t}+n} + \sigma^2 \boldsymbol{U}_{it}^{(n,m)}\right)}{\Phi_{n_{i\bullet t}}\left(\mu_{it}; \boldsymbol{I}_{n_{i\bullet t}} + \sigma^2 \boldsymbol{U}_{it}\right)}.
\end{aligned}$$

ii. The one-step-ahead prediction of the total claim count is

$$\mathbb{E}\left[N_{i,t+1} \mid \mathcal{H}_{it}\right] = \lambda_{i,t+1} \frac{r + n_{i\bullet t}}{r + \lambda_{i\bullet t}} = \lambda_{i,t+1} \left\{\zeta_{it} \frac{n_{i\bullet t}}{\lambda_{i\bullet t}} + (1 - \zeta_{it})\right\}, \qquad \zeta_{it} := \frac{\lambda_{i\bullet t}}{r + \lambda_{i\bullet t}}.$$

iii. The predictive probability that a claim in period $t+1$ is extreme is

$$\widehat{p}_{i,t+1} := \mathbb{E}\left[p_{i,t+1}\left(\Theta_i^{(2)}\right) \mid \mathcal{H}_{it}\right] = \frac{\Phi_{n_{i\bullet t}+1}\left(\mu_{it}^{(1,1)}; \boldsymbol{I}_{n_{i\bullet t}+1} + \sigma^2 \boldsymbol{U}_{it}^{(1,1)}\right)}{\Phi_{n_{i\bullet t}}\left(\mu_{it}; \boldsymbol{I}_{n_{i\bullet t}} + \sigma^2 \boldsymbol{U}_{it}\right)}.$$

iv. For $n \in \mathbb{N}_0$, the one-step-ahead prediction of the extreme claim count given the total claim count is

$$\mathbb{E}\left[M_{i,t+1} \mid \mathcal{H}_{it}, N_{i,t+1} = n\right] = n\, \widehat{p}_{i,t+1}.$$

v. The one-step-ahead prediction of the extreme claim count is

$$\mathbb{E}\left[M_{i,t+1} \mid \mathcal{H}_{it}\right] = \lambda_{i,t+1} \frac{r + n_{i\bullet t}}{r + \lambda_{i\bullet t}}\, \widehat{p}_{i,t+1}.$$

Proof. We first show part i. Applying Theorem 1 to the augmented history $\mathcal{H}_{it} \cup \{(N_{i,t+1}, M_{i,t+1}) = (n, m)\}$ and to $\mathcal{H}_{it}$, we obtain

$$\mathbb{P}\left(N_{i,t+1} = n,\, M_{i,t+1} = m \mid \mathcal{H}_{it}\right) = \frac{L_{i,t+1}^{(1)}}{L_{it}^{(1)}} \cdot \frac{L_{i,t+1}^{(2)}}{L_{it}^{(2)}},$$

where $L_{i,t+1}^{(1)}$ and $L_{i,t+1}^{(2)}$ are evaluated at the augmented history. By (3.3),

$$\frac{L_{i,t+1}^{(1)}}{L_{it}^{(1)}} = \frac{\Gamma\left(r + n_{i\bullet t} + n\right)}{\Gamma\left(r + n_{i\bullet t}\right) n!} \frac{\lambda_{i,t+1}^{n} \left(r + \lambda_{i\bullet t}\right)^{r+n_{i\bullet t}}}{\left(r + \lambda_{i\bullet t} + \lambda_{i,t+1}\right)^{r+n_{i\bullet t}+n}},$$

and, by the last line of (3.4),

$$\frac{L_{i,t+1}^{(2)}}{L_{it}^{(2)}} = \binom{n}{m} \frac{\Phi_{n_{i\bullet t}+n}\left(\mu_{it}^{(n,m)}; \boldsymbol{I}_{n_{i\bullet t}+n} + \sigma^2 \boldsymbol{U}_{it}^{(n,m)}\right)}{\Phi_{n_{i\bullet t}}\left(\mu_{it}; \boldsymbol{I}_{n_{i\bullet t}} + \sigma^2 \boldsymbol{U}_{it}\right)},$$

which proves part i.

To show part ii, note that, conditional on $\left(\Theta_i^{(1)}, \Theta_i^{(2)}\right)$, $N_{i,t+1}$ is independent of $\mathcal{H}_{it}$. Hence, by part ii of Corollary 1,

$$\mathbb{E}\left[N_{i,t+1} \mid \mathcal{H}_{it}\right] = \lambda_{i,t+1} \mathbb{E}\left[\Theta_i^{(1)} \mid \mathcal{H}_{it}\right] = \lambda_{i,t+1} \frac{r + n_{i\bullet t}}{r + \lambda_{i\bullet t}},$$

and the second expression follows from $(r + n_{i\bullet t})/(r + \lambda_{i\bullet t}) = \zeta_{it}\, n_{i\bullet t}/\lambda_{i\bullet t} + 1 - \zeta_{it}$.

To show part iii, part iii of Corollary 1 gives

$$\widehat{p}_{i,t+1} = \frac{\mathbb{E}\left[p_{i,t+1}\left(\Theta_i^{(2)}\right) \prod_{k=1}^{n_{i\bullet t}} \Phi\left(u_k\left(h_k + \Theta_i^{(2)}\right)\right)\right]}{\Phi_{n_{i\bullet t}}\left(\mu_{it}; \boldsymbol{I}_{n_{i\bullet t}} + \sigma^2 \boldsymbol{U}_{it}\right)}.$$

Since $p_{i,t+1}(\theta) = \Phi\left(1 \cdot (\eta_{i,t+1} + \theta)\right)$, the product in the numerator is the product in (3.1) for the history augmented by $(n, m) = (1, 1)$. Applying Lemma 2 to the augmented history proves part iii.

To show part iv, we first observe that, conditional on $\mathcal{H}_{it}$, the joint density of $\left(\Theta_i^{(1)}, \Theta_i^{(2)}\right)$ and the probability mass function of $N_{i,t+1}$ satisfy

$$\pi(\theta_1, \theta_2, n \mid \mathcal{H}_{it}) = \pi(\theta_1 \mid \mathcal{H}_{it}) \mathbb{P}\left(N_{i,t+1} = n \mid \Theta_i^{(1)} = \theta_1\right) \pi(\theta_2 \mid \mathcal{H}_{it}),$$

by part i of Corollary 1 and Model 1. Hence $\Theta_i^{(2)}$ is independent of $N_{i,t+1}$ conditional on $\mathcal{H}_{it}$. Together with the binomial observation equation, this gives

$$\mathbb{E}\left[M_{i,t+1} \mid \mathcal{H}_{it}, N_{i,t+1} = n\right] = n\, \mathbb{E}\left[p_{i,t+1}\left(\Theta_i^{(2)}\right) \mid \mathcal{H}_{it}, N_{i,t+1} = n\right] = n\, \widehat{p}_{i,t+1}.$$

Finally, part v follows from part iv and part ii, since

$$\mathbb{E}\left[M_{i,t+1} \mid \mathcal{H}_{it}\right] = \mathbb{E}\left[N_{i,t+1} \mid \mathcal{H}_{it}\right] \widehat{p}_{i,t+1}.$$

□

Remark 6 (Interpretation of the predictions)**.** *Part ii of Theorem 2 is the exact credibility formula of the Poisson-gamma model: the a priori frequency $\lambda_{i,t+1}$ is multiplied by a credibility-weighted average of the observed ratio $n_{i\bullet t}/\lambda_{i\bullet t}$ and the prior mean 1, with credibility factor ζ_{it}, which increases to one as $\lambda_{i\bullet t} \to \infty$. Part iii is the posterior predictive probability that an additional Bernoulli-probit trial, namely a new claim, is a success. In particular, if $n_{i\bullet t} = 0$, then $\widehat{p}_{i,t+1} = \Phi\left(\eta_{i,t+1}/\sqrt{1+\sigma^2}\right)$, which is the a priori probability that a claim is extreme. Part v shows that the prediction of the extreme claim count is the product of the two exact Bayesian predictions, one*

for the frequency and one for the proportion of extreme claims, as a consequence of the a posteriori independence of the random effects.

We conclude this section with the unconditional moments of Model 1, which describe the dependence induced by the random effects.

Proposition 1 (Moments)**.** *Under Model 1, fix a region i, and define*

$$\pi_{it} := \Phi\left(\frac{\eta_{it}}{\sqrt{1+\sigma^2}}\right), \qquad \pi_{i,st} := \Phi_2\left((\eta_{is},\eta_{it})^\top; \boldsymbol{I}_2 + \sigma^2 \boldsymbol{J}_2\right), \qquad s,t \in \mathbb{N},$$

where $\boldsymbol{J}_2$ is the 2×2 matrix of ones. Then, for $s,t \in \mathbb{N}$ with $s \neq t$, we have

$$\begin{aligned}
\mathbb{E}\left[N_{it}\right] &= \lambda_{it}, & \mathbb{E}\left[M_{it}\right] &= \lambda_{it}\pi_{it},\\
\mathrm{Var}\left(N_{it}\right) &= \lambda_{it} + \frac{\lambda_{it}^2}{r}, & \mathrm{Cov}\left(N_{is},N_{it}\right) &= \frac{\lambda_{is}\lambda_{it}}{r},\\
\mathrm{Cov}\left(N_{it},M_{it}\right) &= \pi_{it}\left(\lambda_{it} + \frac{\lambda_{it}^2}{r}\right), & \mathrm{Cov}\left(N_{is},M_{it}\right) &= \frac{\lambda_{is}\lambda_{it}\pi_{it}}{r},
\end{aligned}$$

and

$$\begin{aligned}
\mathrm{Var}\left(M_{it}\right) &= \lambda_{it}\pi_{it} + \lambda_{it}^2\left\{\left(1+\frac{1}{r}\right)\pi_{i,tt} - \pi_{it}^2\right\},\\
\mathrm{Cov}\left(M_{is},M_{it}\right) &= \lambda_{is}\lambda_{it}\left\{\left(1+\frac{1}{r}\right)\pi_{i,st} - \pi_{is}\pi_{it}\right\}.
\end{aligned}$$

Proof. Write $\Theta_1 := \Theta_i^{(1)}$ and $\Theta_2 := \Theta_i^{(2)}$, and recall $\mathbb{E}\left[\Theta_1\right] = 1$ and $\mathbb{E}\left[\Theta_1^2\right] = 1 + 1/r$. By Lemma 2 with $k=1$ and $k=2$, and $u_{1:k}$ the vector of ones,

$$\mathbb{E}\left[p_{it}(\Theta_2)\right] = \pi_{it}, \qquad \mathbb{E}\left[p_{is}(\Theta_2)p_{it}(\Theta_2)\right] = \pi_{i,st}, \qquad s,t \in \mathbb{N}.$$

Conditional on (Θ_1,Θ_2), Model 1 and Lemma 3 give, for $s \neq t$,

$$\begin{aligned}
\mathbb{E}\left[N_{it} \mid \Theta_1,\Theta_2\right] &= \lambda_{it}\Theta_1, & \mathbb{E}\left[M_{it} \mid \Theta_1,\Theta_2\right] &= \lambda_{it}\Theta_1 p_{it}(\Theta_2),\\
\mathbb{E}\left[N_{it}^2 \mid \Theta_1,\Theta_2\right] &= \lambda_{it}\Theta_1 + \lambda_{it}^2\Theta_1^2, & \mathbb{E}\left[M_{it}^2 \mid \Theta_1,\Theta_2\right] &= \lambda_{it}\Theta_1 p_{it}(\Theta_2) + \lambda_{it}^2\Theta_1^2 p_{it}(\Theta_2)^2,
\end{aligned}$$

and

$$\begin{aligned}
\mathbb{E}\left[N_{it}M_{it} \mid \Theta_1,\Theta_2\right] &= p_{it}(\Theta_2)\mathbb{E}\left[N_{it}^2 \mid \Theta_1,\Theta_2\right] = p_{it}(\Theta_2)\left(\lambda_{it}\Theta_1 + \lambda_{it}^2\Theta_1^2\right),\\
\mathbb{E}\left[N_{is}M_{it} \mid \Theta_1,\Theta_2\right] &= \lambda_{is}\lambda_{it}\Theta_1^2 p_{it}(\Theta_2),\\
\mathbb{E}\left[M_{is}M_{it} \mid \Theta_1,\Theta_2\right] &= \lambda_{is}\lambda_{it}\Theta_1^2 p_{is}(\Theta_2)p_{it}(\Theta_2).
\end{aligned}$$

Taking expectations, using the independence of Θ_1 and Θ_2, and subtracting the products of the means proves the result. □

4. Application to regional claim counts in Korea

In this section, we fit Model 1 to a regional panel of claim counts and evaluate its one-step-ahead predictions.

4.1. Data

The data set contains annual observations for $I = 17$ regions of Korea over the years 2015–2024. For region i and year t, the total claim count N_{it} and the extreme claim count M_{it} are recorded, and $M_{it} \leq N_{it}$ holds for all observations. The covariate vector x_{it} consists of an intercept and the logarithms of four regional characteristics: the recovery budget lagged by one year (RecvBudget_m1Y), the gross regional domestic product (GRDP), the population, and the area. We use the years 2015–2023 as the training period and hold out 2024 for the evaluation of the predictions. The training period contains 153 region-years, 932 claims, and 200 extreme claims. Since 8 region-years have no claims, 145 region-years contribute to the likelihood of the extreme claim part (Remark 4).

Table 1 reports the annual totals. The total claim count varies moderately over the years, whereas the share of extreme claims varies considerably, from 2.4% in 2015 and 4.1% in 2021 to 50.3% in 2020. We return to this feature in Section 4.4.

Table 1: Annual totals of the total and extreme claim counts over the 17 regions.

Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
$\sum_i N_{it}$	42	74	82	121	101	151	123	99	139	151
$\sum_i M_{it}$	1	20	8	9	12	76	5	26	43	32
Share (%)	2.4	27.0	9.8	7.4	11.9	50.3	4.1	26.3	30.9	21.2

4.2. Estimation

By Remark 4, the maximum likelihood estimators of (α, r) and (β, σ) are obtained separately from $\ell^{(1)}$ and $\ell^{(2)}$. The log-likelihood $\ell^{(1)}$ is available in closed form by (3.3). The log-likelihood $\ell^{(2)}$ is evaluated by the one-dimensional integral in Remark 5 using Gauss–Hermite quadrature with 100 nodes; increasing the number of nodes to 200 changes $\ell^{(2)}$ by less than 10^{-6}, and adaptive numerical integration agrees to within 2×10^{-4}. Both log-likelihoods are maximized by a quasi-Newton method with analytic gradients followed by Newton iterations, and the standard errors are obtained from the observed information matrix. The standard errors of the random effect variances $1/r$ and σ^2 are obtained by the delta method.

As a benchmark, we fit the corresponding models without random effects, namely the Poisson generalized linear model (GLM) for N_{it} and the binomial probit GLM for M_{it} given N_{it}. These correspond to Model 1 with $\Theta_i^{(1)} \equiv 1$ and $\Theta_i^{(2)} \equiv 0$, so that the predictions are $\lambda_{i,t+1}$ and $\Phi(\eta_{i,t+1})$, without experience rating. We consider two specifications of x_{it}: Specification A uses all four covariates, and Specification B omits the recovery budget. The random effect models are referred to as the generalized linear mixed models (GLMMs).

Table 2 reports the estimates of the total claim count part. The gamma random effect is supported by the likelihood ratio test, whose p-value is computed from the boundary-corrected $\frac{1}{2}\chi^2_1$ distribution, and by both information criteria. The recovery budget is not significant in either model, and removing it lowers both information criteria.

Table 3 reports the estimates of the extreme claim part. As for the total claim count, the Gaussian random effect is supported by the likelihood ratio test in both specifications, with $\widehat{\sigma}^2 =$

Table 2: Estimates of the total claim count part (α, r), with standard errors in parentheses. GLM: Poisson GLM; GLMM: Poisson-gamma GLMM.

	Specification A		Specification B	
	GLM	GLMM	GLM	GLMM
Intercept	−3.192 (1.024)	−0.096 (2.492)	−2.559 (0.737)	0.399 (1.939)
log(RecvBudget_m1Y)	0.060 (0.067)	0.033 (0.111)	–	–
log(GRDP)	0.313 (0.130)	1.120 (0.432)	0.327 (0.129)	1.179 (0.390)
log(Population)	−0.117 (0.134)	−0.910 (0.415)	−0.115 (0.133)	−0.958 (0.389)
log(Area)	0.247 (0.047)	0.230 (0.088)	0.283 (0.026)	0.247 (0.068)
$1/r$	–	0.091 (0.066)	–	0.100 (0.064)
Log-likelihood	−388.78	−382.69	−389.19	−382.73
AIC	787.55	777.38	786.38	775.47
BIC	802.71	795.56	798.50	790.62
LR test (p-value)	12.18 (0.0002)		12.91 (0.0002)	

0.117 in Specification A and $\widehat{\sigma}^2 = 0.058$ in Specification B, and the GLMM attains the lower AIC and BIC. The recovery budget is again not significant, and Specification B has the lower AIC and BIC among the GLMMs. In both parts, the coefficients of log(GRDP) and log(Population) have opposite signs and change considerably once the random effect is introduced. These two covariates are strongly correlated across regions, and population varies little within a region over time, so that only their contrast, a GRDP per capita effect, is well identified. The coefficient of log(Area) is the most stable across models; a larger area is associated with a higher frequency and a higher share of extreme claims. These associations are descriptive and should not be interpreted as causal effects.

Table 3: Estimates of the extreme claim part (β, σ), with standard errors in parentheses.

	Specification A		Specification B	
	Probit GLM	Probit GLMM	Probit GLM	Probit GLMM
Intercept	−0.220 (1.735)	3.216 (3.269)	0.246 (1.105)	−0.098 (1.808)
log(RecvBudget_m1Y)	0.039 (0.111)	−0.294 (0.206)	–	–
log(GRDP)	0.521 (0.203)	1.033 (0.448)	0.531 (0.201)	0.733 (0.305)
log(Population)	−0.650 (0.209)	−0.968 (0.420)	−0.652 (0.208)	−0.779 (0.316)
log(Area)	0.261 (0.077)	0.444 (0.148)	0.284 (0.044)	0.269 (0.063)
σ^2	–	0.117 (0.088)	–	0.058 (0.041)
Log-likelihood	−310.05	−305.34	−310.12	−306.63
AIC	630.10	622.67	628.23	623.25
BIC	644.98	640.53	640.14	638.14
LR test (p-value)	9.43 (0.0011)		6.98 (0.0041)	
Pearson χ^2/df	4.32	–	4.27	–

4.3. One-step-ahead prediction of 2024

Using the training years as the history $\mathcal{H}_{i,2023}$, we compute the predictive probability $\widehat{p}_{i,2024}$ of an extreme claim from part iii of Theorem 2, and predict $M_{i,2024}$ in two ways:

(a) given the realized total claim count, by $N_{i,2024}\,\widehat{p}_{i,2024}$ (part iv of Theorem 2);

(b) without it, by $\mathbb{E}\left[N_{i,2024} \mid \mathcal{H}_{i,2023}\right] \widehat{p}_{i,2024}$ (part v of Theorem 2), where the first factor is given by part ii of Theorem 2 with the estimates in Table 2.

For the GLM benchmark, $\widehat{p}_{i,2024}$ and $\mathbb{E}\left[N_{i,2024} \mid \mathcal{H}_{i,2023}\right]$ are replaced by $\Phi(\eta_{i,2024})$ and $\lambda_{i,2024}$, respectively, with the GLM estimates. The predictions are evaluated by the root mean squared error (RMSE), the mean absolute error (MAE), and the deviance, which is the binomial deviance in case (a) and the Poisson deviance in case (b).

Table 4 summarizes the results. In contrast to the in-sample comparison, the GLMM does not improve the out-of-sample predictions of the extreme claim counts: the GLM attains slightly lower values of all three criteria in both cases and both specifications. In case (b), the GLMM predicts a total of 33.8 and 34.6 extreme claims, which is closer to the realized total of 32 than the GLM totals of 28.4 and 27.8; the improvement in the total stems from the total claim count part, for which the GLMM predicts the realized total of 151 claims better than the GLM (133.4 and 133.9 against 113.0 and 111.6).

Table 4: Prediction criteria for the extreme claim counts in 2024 (17 regions, realized total $\sum_i M_{i,2024} = 32$).

		Specification A		Specification B	
	Criterion	GLM	GLMM	GLM	GLMM
(a) given $N_{i,2024}$	RMSE	2.341	2.524	2.323	2.569
	MAE	1.556	1.762	1.534	1.786
	Binomial deviance	37.10	43.30	36.66	43.83
	Predicted total	39.7	41.1	39.4	41.7
(b) unconditional	RMSE	1.655	1.780	1.634	1.790
	MAE	1.170	1.304	1.157	1.326
	Poisson deviance	21.74	24.09	21.14	24.50
	Predicted total	28.4	33.8	27.8	34.6

Table 5 reports the region-level results for Specification B, ordered by the posterior mean of $\Theta_i^{(2)}$. The posterior means range from -0.30 for Chungnam to 0.28 for Gwangju, and the credibility adjustment moves $\widehat{p}_{i,2024}$ away from the a priori probability $\Phi\left(\eta_{i,2024}/\sqrt{1+\widehat{\sigma}^2}\right)$ by up to 11 percentage points. The largest prediction errors of the GLMM occur in regions whose share of extreme claims in 2024 departs from their own history. For example, Gyeonggi and Chungnam had 16 of 119 and 14 of 70 claims classified as extreme during the training period, which lowers their predictive probabilities, but 7 of 13 and 7 of 12 claims in 2024. Conversely, Jeonnam and Gyeongnam had high shares during the training period but only 4 of 17 and 2 of 19 in 2024.

4.4. Discussion

The results indicate that the static random effects of Model 1 capture persistent differences between regions, as shown by the likelihood ratio tests, but that the share of extreme claims is dominated by shocks that are common within a year. Two diagnostics support this interpretation. First, the Pearson statistic of the probit GLM in Table 3 is about 4.3 times its degrees of freedom, so the extreme claim counts are strongly overdispersed relative to the binomial distribution. Second,

Table 5: Region-level results for Specification B: training totals, posterior mean of $\Theta_i^{(2)}$, a priori and predictive probabilities of an extreme claim, and realized and predicted extreme claim counts in 2024 given $N_{i,2024}$ (case (a)).

	2015–2023			Probability of extreme claim			2024		
Region	$\sum_t N_{it}$	$\sum_t M_{it}$	$\mathbb{E}\left[\Theta_i^{(2)} \mid \mathcal{H}\right]$	A priori	GLM	GLMM	$N_{i,2024}$	$M_{i,2024}$	GLMM pred.
Gwangju	28	5	0.278	0.079	0.076	0.125	5	0	0.62
Jeonnam	58	25	0.214	0.383	0.367	0.463	17	4	7.87
Busan	46	5	0.165	0.085	0.075	0.110	5	0	0.55
Jeonbuk	68	18	0.162	0.241	0.249	0.289	8	4	2.31
Seoul	35	6	0.145	0.132	0.086	0.161	2	0	0.32
Gyeongnam	64	19	0.129	0.301	0.284	0.344	19	2	6.53
Gyeongbuk	94	38	0.129	0.393	0.376	0.440	9	3	3.96
Chungbuk	70	22	0.067	0.333	0.317	0.354	10	3	3.54
Daejeon	22	1	−0.030	0.082	0.079	0.077	6	1	0.46
Gangwon	82	21	−0.035	0.333	0.345	0.317	22	1	6.97
Daegu	22	1	−0.036	0.097	0.097	0.090	6	0	0.54
Sejong	20	1	−0.067	0.100	0.110	0.088	2	0	0.18
Ulsan	32	4	−0.144	0.260	0.220	0.214	6	0	1.28
Jeju	45	3	−0.146	0.155	0.170	0.121	0	0	0.00
Incheon	57	1	−0.252	0.115	0.101	0.072	9	0	0.65
Gyeonggi	119	16	−0.282	0.280	0.227	0.191	13	7	2.48
Chungnam	70	14	−0.298	0.392	0.354	0.283	12	7	3.39

adding year fixed effects to the probit GLM of Specification B reduces the deviance by 156.9 on 8 degrees of freedom, whereas adding region fixed effects on top of them reduces it by a further 41.0 on 16 degrees of freedom. Even with both sets of fixed effects, the Pearson statistic remains about 3.9 times its degrees of freedom. This suggests that the claims within a region-year are not conditionally independent, as would be the case when several claims arise from the same event. Under such shocks, the experience of a region over the training period is only weakly informative about its share of extreme claims in the next year, which explains why the credibility adjustment of $\widehat{p}_{i,2024}$ does not improve the predictions.

These findings are subject to two caveats. The evaluation is based on a single hold-out year with 17 regions and 32 extreme claims, so the differences in Table 4 are small relative to their sampling variability; a rolling-origin evaluation over several years would give a more reliable comparison. Moreover, the conclusions concern the static specification. Natural extensions of Model 1 are a common year effect, a random effect at the region-year level to absorb the dependence among claims from the same event, and dynamic random effects that discount older experience.

5. Application to the Wisconsin Local Government Property Insurance Fund

In this section, we apply Model 1 to a second data set, in which the units are insured entities rather than regions, the panel is much larger, and the nested count is defined by the type of coverage rather than by the claim size.

5.1. Data

The Wisconsin Local Government Property Insurance Fund (LGPIF) insures local government entities, such as cities, counties, schools, towns, and villages, against property losses. We use the data of E. W. Frees, which record for each entity and year the claim counts of six coverage types: building and contents (BC), inland marine (IM), and four types of vehicle coverage, namely new and old vehicle collision (PN, PO) and new and old vehicle comprehensive (CN, CO). The training data cover 5,677 entity-years of 1,234 entities over 2006–2010, and the hold-out data cover 1,098 entities in 2011, of which all but 9 appear in the training data. We define

$$N_{it} := \text{number of claims over all six coverages},$$
$$M_{it} := \text{number of vehicle claims (PN, PO, CN, CO)},$$

so that $M_{it} \leq N_{it}$ by construction, and M_{it}/N_{it} is the share of vehicle claims among all claims of entity i in year t. Table 6 reports the annual totals. In contrast to the application in Section 4, the share of vehicle claims is stable over the years, between 33% and 39%.

Table 6: Annual totals of the LGPIF claim counts. The year 2011 is the hold-out year.

Year	2006	2007	2008	2009	2010	2011
Entities	1,162	1,147	1,134	1,117	1,117	1,098
$\sum_i N_{it}$	1,415	1,676	1,471	1,675	1,962	1,733
$\sum_i M_{it}$	518	554	566	569	682	612
Share (%)	36.6	33.1	38.5	34.0	34.8	35.3

The covariate vector x_{it} consists of an intercept, indicators of the entity type with cities as the baseline, the indicator IsRC provided in the data, the logarithm of the BC deductible, and $\log(1 + c)$ for the BC, IM, and total vehicle coverage amounts c, in millions. We do not use the no-claim credit variables in the data, because they are themselves based on the claim experience of the entity, which is accounted for by the random effects.

Most entities have no vehicle coverage: 62% of the entity-years have zero vehicle coverage amount, so that the probability of a vehicle claim is zero by construction. We therefore apply the binomial part of Model 1 only to the entity-years with vehicle coverage, and set the predicted vehicle claim count to zero for the entities without it. Only 4 entity-years in the training data, with 9 claims in total, record vehicle claims without vehicle coverage; they are excluded from the binomial part. The binomial part is therefore estimated from 1,110 entity-years of 391 entities with vehicle coverage and at least one claim, which contain 6,294 claims, of which 2,880 are vehicle claims.

5.2. Estimation

The estimation follows Section 4.2, with the entity in place of the region. For the binomial part, the Gauss–Hermite quadrature uses 200 nodes. Since some entities have several hundred claims

over the training period, their likelihood contributions are sharply peaked in θ; with 200 nodes, $\ell^{(2)}$ agrees with that computed by 400 nodes and by adaptive numerical integration to within 5×10^{-5}.

Table 7 reports the estimates. Both random effects are strongly supported. For the total claim count, the variance of the gamma random effect is $1/\widehat{r} = 0.735$, the Pearson statistic of the Poisson GLM is 5.18 times its degrees of freedom, and the log-likelihood increases by 3,176. For the vehicle claims, the variance of the Gaussian random effect is $\widehat{\sigma}^2 = 0.360$, the Pearson statistic of the probit GLM is 2.15 times its degrees of freedom, and the log-likelihood increases by 395. Both likelihood ratio tests have p-values below 10^{-100}. Several coefficients change markedly once the random effects are introduced, and their GLM standard errors are considerably smaller than the GLMM ones, which reflects the neglected dependence among the repeated observations of an entity in the GLMs.

Table 7: Estimates for the LGPIF data, with standard errors in parentheses. Total claims: Poisson GLM and Poisson-gamma GLMM; vehicle claims: probit GLM and probit GLMM.

	Total claims N_{it}		Vehicle claims M_{it}	
	GLM	GLMM	GLM	GLMM
Intercept	−1.611 (0.065)	−1.093 (0.179)	1.090 (0.106)	−0.548 (0.261)
TypeCounty	0.791 (0.037)	0.478 (0.170)	0.915 (0.066)	0.831 (0.172)
TypeMisc	−1.463 (0.097)	−0.648 (0.167)	0.236 (0.277)	0.152 (0.369)
TypeSchool	−0.655 (0.050)	−0.486 (0.128)	0.496 (0.105)	0.405 (0.204)
TypeTown	−0.225 (0.094)	−0.425 (0.156)	−0.998 (0.176)	−0.610 (0.267)
TypeVillage	0.006 (0.049)	0.014 (0.115)	−0.263 (0.095)	−0.097 (0.176)
IsRC	−0.343 (0.025)	−0.036 (0.073)	0.011 (0.037)	−0.133 (0.089)
log(1 + BC coverage)	0.856 (0.017)	0.702 (0.040)	−0.536 (0.028)	−0.350 (0.062)
log(BC deductible)	−0.121 (0.009)	−0.191 (0.024)	−0.005 (0.014)	0.137 (0.034)
log(1 + IM coverage)	−0.399 (0.031)	0.198 (0.096)	0.017 (0.052)	−0.068 (0.142)
log(1 + vehicle coverage)	0.516 (0.015)	0.487 (0.052)	0.350 (0.041)	0.377 (0.086)
$1/r$ or σ^2	–	0.735 (0.048)	–	0.360 (0.053)
Observations	5,677		1,110	
Log-likelihood	−9,160.5	−5,984.2	−1,700.8	−1,305.3
AIC	18,342.9	11,992.3	3,423.6	2,634.7
BIC	18,416.0	12,072.1	3,478.7	2,694.8
Pearson χ^2/df	5.18	–	2.15	–

The estimates are consistent with the coverage structure: a larger vehicle coverage is associated with more claims in total and with a higher share of vehicle claims, whereas a larger BC coverage is associated with more claims in total but a lower share of vehicle claims, since it increases the number of BC claims. Counties have both more claims and a higher share of vehicle claims than cities, and towns have a lower share. As in Section 4, these associations are descriptive.

5.3. One-step-ahead prediction of 2011

We predict the claim counts of 2011 from the training history as in Section 4.3: the total claim count by $\mathbb{E}\left[N_{i,2011} \mid \mathcal{H}_{i,2010}\right]$, and the vehicle claim count (a) given the realized total claim count by $N_{i,2011}\,\widehat{p}_{i,2011}$, and (b) without it by $\mathbb{E}\left[N_{i,2011} \mid \mathcal{H}_{i,2010}\right]\,\widehat{p}_{i,2011}$. The 9 entities without training

data receive the a priori predictions. Table 8 reports the results.

Table 8: Prediction criteria for 2011 (1,098 entities, of which 405 have vehicle coverage). The binomial deviance is computed over the entities with vehicle coverage and $N_{i,2011} \geq 1$, and the Poisson deviance of M over the entities with vehicle coverage.

	Criterion	GLM	GLMM
Total claims N	RMSE	7.401	3.563
	MAE	1.473	0.941
	Poisson deviance	2,902.7	1,230.8
	Predicted total (actual 1,733)	1,780.2	1,880.2
Vehicle claims M, (a) given $N_{i,2011}$	RMSE	0.898	0.488
	MAE	0.212	0.142
	Binomial deviance	426.0	270.4
	Predicted total (actual 612)	627.3	646.0
Vehicle claims M, (b) unconditional	RMSE	2.137	1.208
	MAE	0.490	0.290
	Poisson deviance	792.2	343.2
	Predicted total (actual 612)	598.2	693.2

In this application, the GLMM improves all entity-level criteria substantially. For the vehicle claims given $N_{i,2011}$, the RMSE decreases from 0.898 to 0.488 and the binomial deviance from 426.0 to 270.4; without $N_{i,2011}$, the RMSE decreases from 2.137 to 1.208. The credibility adjustments are large: the posterior means $\mathbb{E}\left[\Theta_i^{(1)} \mid \mathcal{H}_{i,2010}\right]$ range from 0.05 to 27.3 with median 0.80, and the posterior means of $\Theta_i^{(2)}$ range from -1.92 to 1.31 among the entities with vehicle coverage. The improvement is not driven by a few large entities. Excluding the 11 entities in the top 1% of $N_{i,2011}$, the RMSEs of the GLM and the GLMM are 2.57 and 1.89 for N, and 0.92 and 0.65 for M given $N_{i,2011}$. In 2,000 bootstrap resamples of the entities, the GLMM has the smaller RMSE in 97.0% of the resamples for N and in all resamples for M, in both cases (a) and (b). The GLMMs over-predict the portfolio totals, by 8.5% for N and 13.3% for M in case (b), whereas the GLMs predict the totals more closely; the entity-level accuracy, which matters for experience rating, is nevertheless much higher for the GLMMs.

5.4. Comparison of the two applications

The two applications lead to different conclusions about the benefit of the experience rating of the nested count. In the LGPIF data, the share of vehicle claims of an entity is a persistent characteristic, determined largely by the composition of its insured property, and the annual share in the portfolio is stable. The static random effect $\Theta_i^{(2)}$ captures this persistence, and the credibility predictions of Theorem 2 improve on the GLM substantially. In the regional data of Section 4, the share of extreme claims is dominated by year-specific shocks that are common to the regions, and the experience of a region is only weakly informative about its next-year share. The static model is therefore well suited to nested counts whose split reflects stable unit characteristics, while nested counts driven by common shocks call for the extensions discussed in Section 4.4.